THE UNIVERSITY OF NOTRE DAME AUSTRALIA

MED 4000
WEEK 4.03

TUTOR GUIDE

The cases and materials in this guide were written and developed by;

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The cases and materials in this guide were reviewed by;

- Clinical Years Curriculum Committee, critical care working group. Last reviewed Nov 2010.

RELEVANT PRIOR LEARNING

MED1000

1.09 Pharmacokinetics I, II (L)
1.14 Pharmacokinetics III, IV (L)
1.16 Opiods (L)

MED2000

2.13 Medications for pain relief (L)

MED 3000

3.11   Pain management in malignancy [short case tutorial]

WEEK 03_ANAESTHESIA		
Yr	**Wk**	**LEARNING OBJECTIVES**
4	**3**	• Demonstrate an understanding of the physiology of nausea and vomiting by being able to explain PONV to a patient and outline the mechanism of action of medications used in the management. • Detail the risk factors for post operative nausea and vomiting. (PONV) • Outline preventative steps that can be implemented by an anaesthetist to minimise post operative nausea and vomiting. • Detail a management plan for a patient experiencing post operative nausea and vomiting. • Know the mechanism of action, indications, contraindications and possible adverse effects of the following medications used in the management of post operative nausea and vomiting, ondanestron, desxamethasone, droperidol, metoclopramide, prochlorperazine & promethazine
4	**3**	• Define pain and know the differentiating features of acute and chronic pain. • Know the anatomical pathways of pain perception. • Know the key components of a pain history in the assessment of post operative pain. • Outline the principles of multimodal analgesia in the management of pain. • Know the mechanism of action, indications, contraindications, side effects and dosage of the following medications paracetamol, non-steroidal anti-inflammatory drugs [NSAIDS], oxycodone, codeine and tramadol. • Explain patient controlled analgesia [PCA] and outline the advantages of using PCA in the management of post-operative pain. • Demonstrate an understanding of the assessment and management of post operative pain by being able to outline a pain management plan.
4	**3**	• Outline a holistic pain assessment and management plan for patients with cancer that demonstrates an understanding of the nature of the disease, the common causes of pain, the adverse effects of treatment and patient comfort and needs.
4	**3**	• Outline the key components of the assessment of the post operative patient who is deteriorating. • Demonstrate a systematic approach to the assessment and management of the hypoxaemic post operative patient. • Revise the problem of obstructive sleep apnoea [OSA] and outline how the presence of OSA influences the peri-operative management of the patient.

CASE ONE

Short case number: 4_03_01
Category: Clinical Skills
Discipline: Anaesthesia
Setting: Hospital_short stay surgery ward
Topic: Post Operative Nausea & Vomiting

CASE
Shelly is a 35 year old woman who presents for gynaecological laparoscopy to investigate pelvic pain. She has a history of intermittent asthma, but is otherwise well and uses a salbutamol metered dose inhaler intermittently. She reports a history of severe nausea and vomiting following a general anaesthetic at age 16 years for ORIF fractured ankle. She is concerned that she will experience the same misery again and asks for your advice.

QUESTIONS
1. What are the risk factors for post-operative nausea and vomiting (PONV)? 2. Shelly asks what causes you to feel nauseous and vomit. Revise the physiology of nausea and vomiting in your explanation to Shelly. 3. Shelly appreciates your explanation and asks if anything can be done to prevent it happening again. What preventative steps for PONV can an anaesthetist take? 4. Write short notes on the key classes of drugs that are used in the management of PONV? 5. Despite your preventative measures, Shelly experiences nausea and vomiting in the recovery ward. What additional therapies are available to you to relieve her symptoms?

Resources:

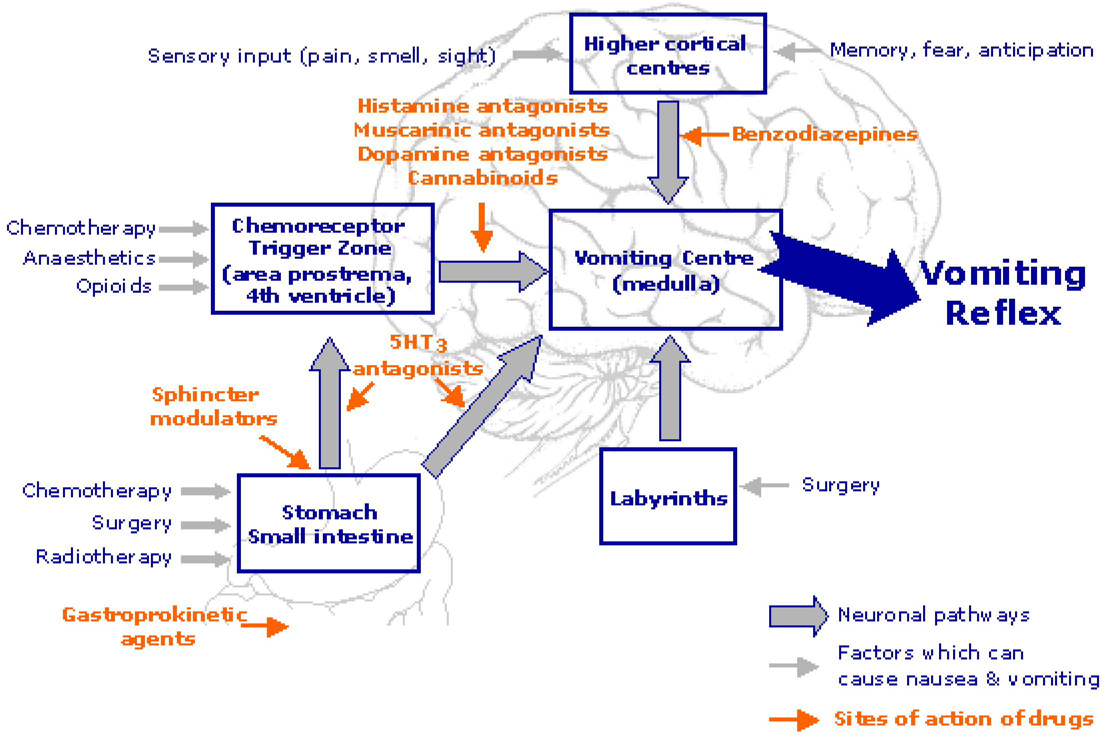
- **Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.**
- **Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.**

Additional Resources:

- **Pleuvery B Physiology and Pharmacology of Nausea and Vomiting Anaesthesia and Intensive Care Medicine 7:12 2006**

ANSWERS

Revise the physiology of nausea and vomiting in your explanation to Shelly.



Many sources of input are integrated via the vomiting centre in the reticular formation of the medulla.

Pharyngeal and gastrointestinal irritation/distension sends afferent impulses via vagus to the CTZ and VC.
Chemoreceptor trigger zone (CTZ), found in the area postrema in the floor of the fourth ventricle, is rich in 5-HT3, dopamine(D2) and opioid receptors. It is functionally outside the BBB.
Vestibular nuclei send input to the vomiting centre
Higher centres, particularly the limbic system, send direct input to the VC to mediate nausea and vomiting induced by sensory-affective triggers and memory associations.

Neurotransmitters/Receptors involved:
Irritation/distension of the GI tract -> 5-HT/5-HT3 R
Toxins/stimulants at the chemoreceptor trigger zone (CTZ) -> dopamine/D2 R + 5-HT/5-HT3 R
Motion on vestibular nuclei -> ACH + histamine/muscarinic Ach R + H1 R
Direct input to the VC -> Ach + histamine/muscarinic Ach R + H1 R

What are the risk factors for post-operative nausea and vomiting (PONV)?

Risk factors for PONV

Pre-existing patient factors

- female>male
- age
- history of PONV, motion sickness or migraine
- obesity
- non-smokers

Type of surgery
- gynaecological
- gastrointestinal
- neurosurgery
- middle ear
- ophthalmic

Anaesthetic/Post-op factors
- emetogenic drugs
- dehydration/hypotension
- early oral intake
- early ambulation

What preventative steps for PONV can an anaesthetist take?

Anxiolysis

Prophylactic therapy (drugs chosen according to risk stratification)
Avoidance of emetogenic drugs:
- regional technique
- volatile anaes agents ie TIVA technique
- N2O (especially high FiO2 in air)
- Opioids
- neostigmine

Intravenous hydration
Careful transfer of patient

Write short notes on the key classes of drugs that are used in the management of PONV?

Antiemetic Drugs

1. Dopamine antagonists

(i) Metoclopramide
actions
- central D2 receptor antagonism in the CTZ/VC
- prokinetic effect to counteract gastroparesis
- some 5-HT3 antagonism at higher doses

Dose 10mg q8h up to 1-2mg/kg
Side effects include EP effects (dystonia. oculogyric crisis), drowsiness, dizziness
Neuroleptic malignant syndrome (NMS) has been reported

(ii)Phenothiazines
central dopamine (D2) blockade in addition to antihistaminergic, antiadrenergic and anticholinergic effects

- prochlorperazine
- chlorpromazine

Side effects include dystonia, dyskinesia, prolonged QT interval, NMS

(iii) Butyrophenones
Droperidol, Haloperidol
Central D2 receptor antagonism in addition to antiadrenergic and anticholinergic effects (similar to phenothiazines)

Dose
Droperidol: 0.25 – 0.5mg q12h IV
Haloperidol: 1-15mg daily in divided doses po, 1-5mg IV
Side effects: sedation, dysphoria, GI disturbance, EP side effects, abnormal LFT's, QT prolongation

3. 5-HT3 receptor antagonists
actions

- block central and peripheral effects of 5-HT3 to prevent initiation of vomiting reflex

Class example:
Ondansetron

- dose: 4-8mg q8h
- side effects: constipation, headache, flushing

4. Antihistamines

Promethazine
Actions

- competitive antagonist of H1 receptors present in VC and vestibular nucleus
- additionalanticholinergic activity probably contributes to antiemetic effect

Dose: up to 25mg q8h
* caution with intravenous administration – highly irritant to veins
complications of IV injection include pain, swelling and thrombophlebitis, extravasation can cause tissue necrosis and inadvertent arterial injection could cause arterial spasm and ischaemia.

Non-sedating antihistamines (terfenadine, astemizole) are not antiemetic as they do not cross the BBB.

5. Steroids
Dexamethasone
Mechanism of action unclear, but evidence shows benefit in both PONV and chemotherapy
Dose: 4-8mg IV once
Side effects: no significant steroid-related side effects for single prophylactic dose

6. Anticholinergic agents

Mechanism of action viablockade of central muscarinic Ach receptors present in VC and vestibular nuclei therefore recognized role in motion sickness and vestibular disorders

- parenteral or transdermal administration
- side effects include sedation, dry mouth

7. Benzodiazepines

Especially lorazepam

Role via anxiolysis and sedation +/- direct antiemetic effect

CASE TWO

Short case number: 4_03_02
Category: Clinical Skills
Discipline: Anaesthesia
Setting: Surgical Ward
Topic: Post Operative pain management.

CASE
Susan is a 72 year old female who had a right total knee replacement (TKR) under general anaesthetic yesterday. You are reviewing her on the acute pain management round. **She is on lisinopril for treatment of hypertension, and had been using Panadeine Forte, 6 tablets per day and Diclofenac PRN prior to surgery.**

QUESTIONS

1. Define pain. What differentiates Acute from Chronic Pain?
2. The consultant asks you to explain the anatomical pathways of pain perception. Briefly outline the major anatomical pathway of pain perception.
3. What are the key components of your history in assessing Susan's pain?
4. Because of Susan's long history of osteoarthritis requiring analgesia a multimodal approach is considered. What is multimodal analgesia?
5. What are the goals of post-operative pain management?
6. Write short notes on the mechanism of action in pain relief, indications, contraindications and side effects of;
   -paracetamol
   -non-steroidal anti-inflammatories (NSAIDS)
   -common oral opioid preparations
7. Following discussion regarding the management of Susan's pain, Patient controlled analgesia is considered. What is Patient Controlled Analgesia (PCA)?
8. Outline an analgesic regimen for Susan that would often be utilised for acute pain management following TKR.

Resources:

- **Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.**
- **Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.**

ANSWERS

Define pain. What differentiates Acute from Chronic Pain?

Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage.

Nociception is the process of signal transduction and transmission from the periphery to central processing areas via the nervous system.

Pain is a multidimensional experience with sensory-discriminative and affective-motivational components. Ie. nociception does not equal pain

Because on the complex nature of pain, assessment and management are best approached using a BIO-PSYCHO-SOCIAL paradigm. This is distinct from the traditional medical model which tends to address only the BIO component.

BIO – pathophysiology, investigations, treatments, side effects etc
PSYCHO – mood, beliefs, impact of pain
SOCIAL - effects on relationships, life role and other events

Nomenclature dictated in part by duration and relationship to cause.

Acute Pain: considered within 3 months of onset, and associated with a specific injury or disease process

Chronic or Persistent Pain: duration greater than 3 months and/or extending beyond the anticipated normal healing period

Briefly outline the major anatomical pathway of pain perception.

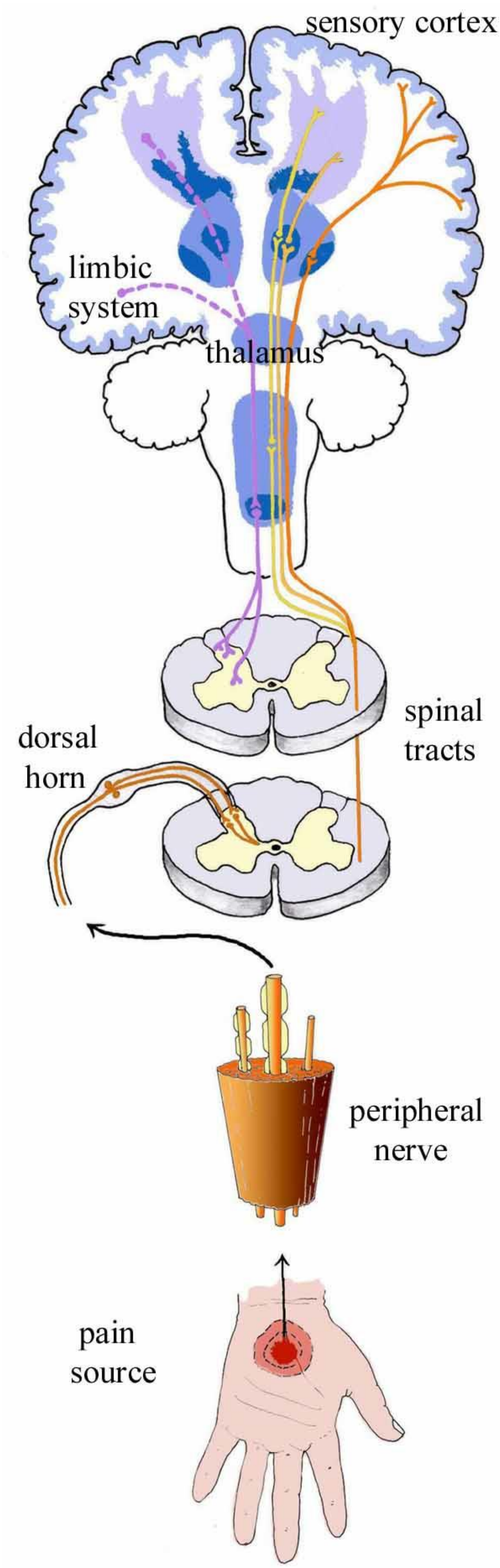
Nociceptors in the skin and other somatic tissues exist as free nerve endings. These are depolarized by mechanical, thermal and chemical noxious stimuli and initiate an electrical stimulus which is transmitted via myelinated Adelta or unmyelinated C fibres (primary afferent nerve fibres).

These primary afferent nerve fibres enter the spinal cord via the dorsal roots and synapse with second order neurons in the dorsal horn of the spinal cord. At this location, there may be further communication with a number of interneurons that further modulate nociceptive input.

From the dorsal horn, nerve fibres project to the midbrain via a number of tracts, the most substantial of which is the spinothalamic tract located anterolaterally in the spinal cord. It terminates in a number of locations that are considered responsible for perception of the various aspects of pain:

Ventral posterolateral thalamus – discriminative aspects such as location, intensity, duration
Medial thalamus – autonomic responses and unpleasant emotional aspects
Reticular formation and hypothalamus – arousal response to pain
Periagueductal grey matter - probable link to descending inhibitory pathways

Projections continue from the thalamus to higher centres in the cerebral cortex to complete the multidimensional perception of pain, including influences on memory, affect and behaviour.


sensory cortex
limbic system
thalamus
spinal tracts
dorsal horn
peripheral nerve
pain source

What are the key components of your history in assessing Susan's pain?

Key Features of a Pain History

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Nature
Intensity
Location and radiation pattern
Duration
Onset
Offset
Concomitant symptoms
Associated symptoms/ Aggrevating factors
Relieving factors
Frequency

In addition:

Treatment history and treatment response
Impact of pain on sleep, activity, mood, work, relationships
Patent beliefs and expectations

What is multimodal analgesia?

This describes the concept of utilizing drugs and other techniques in combination (multiple modes) to provide optimal analgesia with minimal side effects. Underlying its application is the principle of selecting the most appropriate drug/technique to target the mechanism of the pain.
Pharmacological examples include: NSAIDS for inflammatory pain, antiepilieptics for neuropathic pain, conduction block via regional techniques
Non-pharmacological modalities include heat packs, elevation of an inflamed body part, TENS, hypnotherapy and other cognitive techniques, acupunture

What are the goals of post-operative pain management?

(i)Provide analgesia

- in addition to the ethical obligation involved, good analgesia minimises the surgical stress response and has a positive impact on perioperative cardiac risk and emotional well-being

(ii)Minimise adverse effects from analgesic drugs

- using a combination of drugs with different mechanisms of action is aimed at opioid sparing
- in turn, this minimises the opioid related side effects that are the most concerning with regards to patient safety (resp depression, cognitive impairment) and delay in return to function (sedation, nausea/vomiting, ileus, constipation)

(iii) Optimise function

- early mobilisation is critical in prevention of respiratory complications and DVT
- good analgesia facilitates aggressive physiotherapy to ensure optimal return to function. TKR is a good example of this as failure of early active knee flexion is a harbinger of poor functional outcome from the replaced joint

Pharmacology of Common Analgesics

Paracetamol

- analgesic and antipyretic with both peripheral and central actions
- oral, intravenous and PR formulations
- adult dose 500mg to 1gm qid max 4gm per 24hours
- hepatotoxic in overdose (OD)
- potential for hepatotoxicity, particularly in elderly, low body weight or compromised nutrition (relative OD)

NSAIDS

- chemically disparate group of analgesics/antipyretics
- unified by action to inhibit cyclooxygenase activity, hence synthesis of prostaglandins
- group side effects: platelet inhibition, GI mucosal inflammation/erosion/ulceration, renal impairment, hypertension, numerous drug interactions

Oxycodone

- potent oral opioid analgesic
- oral tablet and syrup
- 15mg oral oxycodone equianalgesic to 30mg oral morphine
- group opioid side effects
- minimal active metabolites

Codeine

- low potency opioid analgesic
- 180mg oral codeine equianalgesic to 30mg oral morphine
- group opioid side effects, particularly constipation
- antitussive

Tramadol

- low potency analgesic
- mechanism of action combines opioid and monoaminergic augmentation
- IV and oral formulations
- opioid group side effects in addition to nausea, headache and dysphoria presumably secondary to monoaminergic effects
- drug interactions: TCA's, SSRI's, MAO's

What is Patient Controlled Analgesia (PCA)?

This describes the use of programmable devices for analgesic modalities such as intermittent bolus delivery (+/- infusion) of

(i) intravenous opioid
(ii) epidural infusion solution
(iii) regional nerve catheter infusion solution.

The advantages of these devices compared to intermittent IMI opioid therapy are

(i) enhanced patient autonomy and satisfaction
(ii) reduced nursing workload

(iii) better dynamic analgesia with post-operative incident pain (eg. mobilising with physiotherapist)

Typical Analgesic regimen for TKR

PCA morphine 1-2mg bolus 5min lockout
Paracetamol 1gm qid
NSAID eg diclofenac 50mg tds
Step down to oxycodone combination of sustained and immediate release

Variations include:
Use of intrathecal morphine
Regular oral opioid in lieu of PCA
Epidural
Femoral Nerve sheath catheter
Addition of PPI to pulse NSAID course

CASE THREE

Short case number: 4_03_03
Category: Clinical Skills
Discipline: Anaesthesia
Setting: Ward
Topic: Chronic Pain

CASE
Meredith is a 62y.o. woman with metastatic breast cancer. At diagnosis 12 months ago, she was treated with right breast lumpectomy, axillary node clearance and chemoradiotherapy. Hepatic metastases have subsequently developed. **She has been referred to the Pain Clinic for assessment of progressive right shoulder and arm pain**.

QUESTIONS
1. Describe a framework for assessing pain in cancer patients. 2. What are the common locations of breast cancer metastases? 3. What are the potential causes for Meredith's presentation? 4. Describe a framework for the management of cancer pain.

Resources:

- **Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.**
- **Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.**

ANSWERS

Describe a framework for assessing pain in cancer patients.

Pain and other unpleasant symptoms in cancer patients can originate from:

Disease
- Tissue invasion/destruction, obstruction of hollow viscus
- Pathological fracture
- Paraneoplastic syndromes ie peripheral neuropathy
- Cachexia/deconditioning predisposes to pressure sores

Treatment
- Radiotherapy
  - Acute: tissue inflammation and burns, mucositis
  - Chronic: neuropathy/plexopathy, tissue damage ie. proctitis
- Chemotherapy
  - Acute: mucositis
  - Chronic: neuropathy, GVHD
- Procedures ie. cannulation, lumbar puncture
- Drug side effects ie. constipation, nausea

Unrelated disease processes (tendency to older age group)

Assessment should be undertaken using the BIO-PSYCHO-SOCIAL (+/- SPIRITUAL) paradigm. Due to the potential life shortening nature of illness in cancer patients, attempts should be made to address prognosis, and the patient's understanding of this. This should introduce discussion of end-of-life issues that cross psychololgical, social and spiritual domains in a timely and sensitive fashion.

What are the common locations of breast cancer metastases?

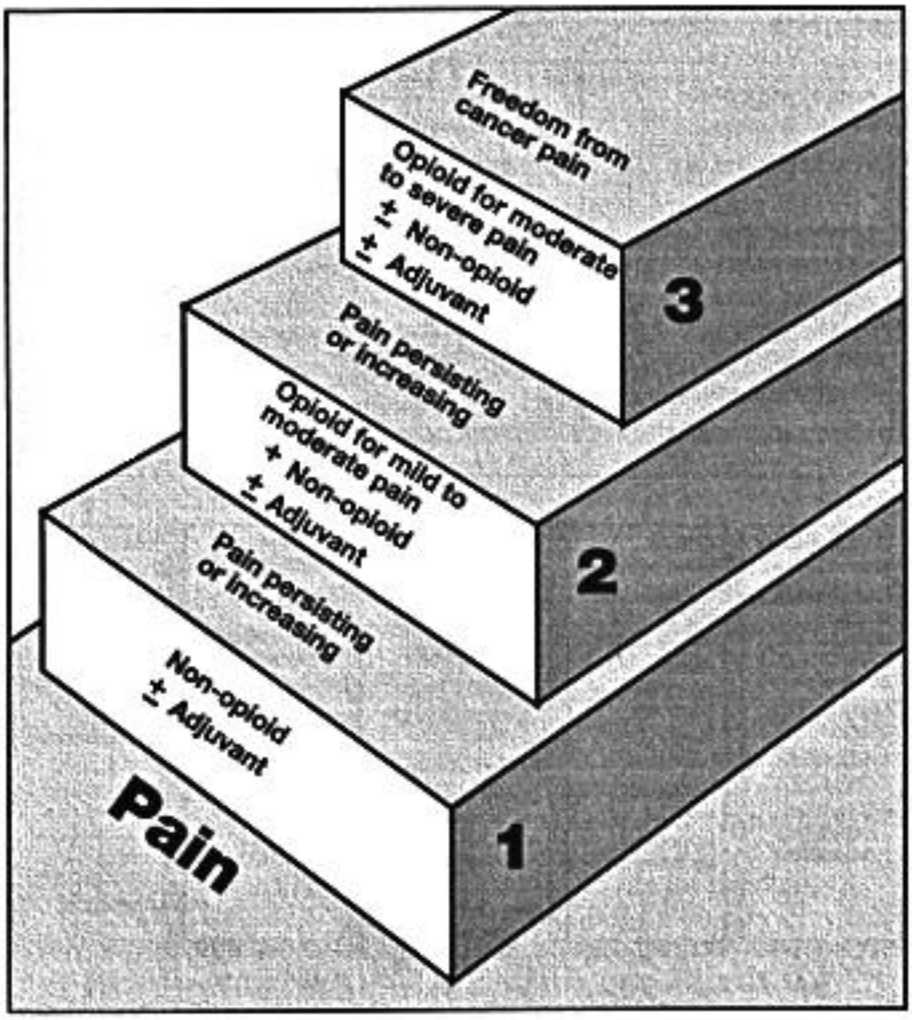
- Bone
- Liver
- Brain
- Adrenals

What are the potential causes for Meredith's presentation?

(i) humeral/pectoral girdle bony metastatic disease
(ii) brachial plexopathy causing neuropathic pain secondary to radiotherapy/+/- chemotherapy or malignant invasion
(iii) unrelated mechanical shoulder pain ie. osteoarthritis glenohumeral joint
(iv) radicular pattern or referred pain from the cervical
  - metastatic disease
  - degenerative disease

Describe a framework for the management of cancer pain.

The WHO three-step analgesic ladder



Source: World Health Organization, 1990. Used with permission.

Management aims are to optimize disease-arresting therapies, and alleviate adverse effect of both disease and treatment. Principles of multimodal, targeted analgesia apply.

To address:

<u>Disease</u>

(i) agents to reduce tumour size or halt disease progression
- curative surgery
- chemo- and radiotherapy

(ii) palliative procedures
- fixation of pathological fractures
- gastrointestinal stent or bypass procedures
- ureteric stents

Pain

(i) Pharmacology
- WHO analgesic ladder
- Differentiate and target somatic/visceral and neuropathic pain

(ii) Interventions
- Neurolytic blocks: coeliac plexus block, intrathecal phenol
- Neuraxialopioid and adjuvant delivery: intrathecal or intraventricular systems

(iii) Physical therapy
- heat/cold packs
- mobility aids
- maintenance of condition

(iv) Psychological interventions
- manage mood disorder
- address grief/loss (include family)
- "get affairs in order"

Adverse Effects of Treatment

(i) antiemetics

(ii) constipation management

(iii) minimization of sedation
- appropriate opioid therapy
- considerneurolytic pain interventions or intrathecal/intraventricular opioid delivery
- stimulants to counteract sedation

Comfort measures

(i) antisialogogues

(ii) oxygen therapy to relieve dyspnoea

(iii) terminal sedation

(iv) religious ceremonies eg. last rights

CASE FOUR

Short case number: 4_03_04
Category: Clinical Skills
Discipline: Anaesthesia
Setting: Recovery
Topic: Post operative monitoring_location of care

CASE
Sally is a 56y.o. registered nurse. She has just undergone revision of her gastric fundoplication. Unfortunately, the procedure had to be converted from laparoscopic approach to open approach due to bleeding and technical difficulties. **She was extubated and transferred to recovery with PCA morphine for analgesia. The recovery room nurse asks you to review her because she requiring high-flow oxygen to maintain oxygen saturations of 92% and appears quite drowsy.** **She weighs 96kg and is160cm tall. She is lying supine with head elevation of 15 degrees. Other medical history includes: gastro-oesophageal reflux disease with oesophageal ulceration and Barrett's oesophagus, hypertension, type 2 diabetes and obstructive sleep apnoea.** **You review the anaesthetic record and discover:** - **that she had a grade 3 Cormack and Lehane view at laryngoscopy, and a bougie was used to facilitate intubation** - **analgesia intraoperatively was morphine 12mg** - **suxamethonium followed by rocuronium was used for muscle relaxation during the 2 hour case**

QUESTIONS
1. Describe your initial approach to this patient's care. 2. What issues are identified that require consideration in her ongoing management? 3. What are the potential causes of hypoxaemia in the post-operative setting? Highlight the causes relevant to Sally and outline a management plan to address these. 4. How does obesity impact on perioperative management? How is obesity categorized? 5. What is obstructive sleep apnoea and how does it influence perioperative management? 6. What is the appropriate location of care following discharge from recovery for this woman?

Resources:

- **Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.**
- **Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.**

ANSWERS

Describe your initial approach to this patient's care.

Students should recognize the suboptimal oxygenation described in this scenario, particularly in light of the patients' difficult airway and multiple co-morbidities.

Review the patient to make a comprehensive assessment including attention to ABC, in addition to history, examination and relevant investigations.

What issues are identified that require consideration in her ongoing management?

Patient:

Hypoxaemia in PACU (*recognize suboptimal oxygenation)
Pre-existing obesity, OSA predisposing to upper airway obstruction and possible central contributors to disordered ventilation.
Semirecumbent position contributing to reduction in FRC
Difficult intubation

Surgery:

Upper GI surgery (laparoscopic, then open) with consequent impairment of respiratory mechanics (ie. diaphragmatic splinting, reduction of FRC, basal atelectasis)

Anaesthetic:

Opioid-dependent analgesic regime, absence of multimodal approach
Muscle relaxant used -> potential for residual NMJ blockade impairing respiratory effort

What are the potential causes of hypoxaemia in the post-operative setting? Highlight the causes relevant to Sally and outline a management plan to address theses.

Classification of the causes of hypoxaemia can be described in 5 functional categories. In the perioperative setting, the cause of hypoxaemia is commonly multifactorial.

Reduced FiO_2

- effective outcome of diffusion hypoxia with N_2O (reduced alveolar FiO_2)
- failure to connect oxygen tubing to supply with transfer

(*diffusion hypoxia: N_2O washout can decrease alveolar O_2 fraction)

Hypoventilation

- airway obstruction: oropharyngeal, laryngospasm, bronchospasm, foreign body
- impaired ventilatory drive: respiratory depressant drugs eg. opioids, anaesthetic agents, residual NMJ blockade, OSA, intracranial pathology
- restriction of diaphragmatic excursion: pain (especially upper GI and thoracic surgery/trauma), obesity, abdominal distension
- presence of air or fluid in the pleural cavity: PTX, HTX, pleural effusion

Ventilation-Perfusion Abnormalities

- decreased FRC and may be exceeded by closing volume
- cardiac output and PA pressure may be reduced
- areas with high V/Q ratio represent physiological deadspace
- areas with low V/Q ratio increase venous admixture

Shunt

- refers to blood that enters to arterial system without going passing through ventilated areas of blood (physiological and pathological)
- *of minimal relevance to post-op setting except in patients with intracardiac shunts or pulmonary A-V shunts

Impaired Diffusion

- pulmonary interstitial oedema: impaired LVF, excess IV fluid infusion, post-obstructive pulmonary oedema, TRALI, pulmonary contusion

relevant to Sally:

- hypoventilation:
    - maximize FiO2
    - airway obstruction – jaw thrust
    - sit upright to optimize resp mechanics
    - opioids - consider trial of naloxone
    - non-opioid analgesia ie. paracetamol, NSAIDs to opioid-spare
    - NMJ blockade – reversal dose +/- repeat dose

- V/Q abnormality
    - optimise ventilatory effort
    - encourage deep breathing and coughing -> chest physio in PACU
    - ensure adequate cardiac output and PA pressure

How does obesity impact on perioperative management? How is obesity categorized?

Issues for consideration:

Presence of associated co-morbidities eg. hypertension, IHD, DM, OSA
Special load-bearing equipment required ie. operating tables, trolleys
Manual handling issues
Poor venous access
Monitoring issues ie NIBP
Difficult airway management
Ventilation
Altered pharmacokinetics/dynamics
Positioning and pressure area care
DVT prophylaxis
Post-operative location of care

Body Mass Index (BMI) is a derived figure that adjusts a patient's weight according to their height to give a meaningful and comparable indicator of body habitus.

BMI = weight (kg)/height (metres squared)

BMI	Descriptor
<20	Underweight
20-24.9	Ideal
25-29.9	Overweight
30-39.9	Obese
40- 49.9	Morbidly obese
50- 59.9	Super obese

Sally's BMI is 37.5.

What is obstructive sleep apnoea and how does it influence perioperative management?

Disorders of sleep and respiratory control are common in obese patients.
These fall into 2 main categories that may co-exist:
(i)) obstructive sleep apnoea (OSA)
(ii) obesity hypoventilation syndrome (OHS).

OSA is characterized by intermittent airway obstruction, reduced tidal volume (hypopnea) and apnoea during sleep. These changes result in nocturnal periods of hypoxaemia and hypercarbia and episodes are terminated by sudden arousal. Clinical features include:

Snoring
Nocturnal apnoeas
Daytime somnolence
Morning headache
Mood disturbance
Hypertension: systemic and pulmonary
Cor pulmonale
*blunting of response to hypoxaemia and hypercarbia

OHS is characterized by reduced central respiratory drive and hypercarbia in the awake state. Prevalence rises with increasing BMI, and it is associated with the same cardiovascular changes as in OSA.

Perioperative Management:
* increased risk of perioperative respiratory depression
(i) Preoperative assessment and management
(i) Perioperative use of CPAP and higher level monitoring
(iii) Maximal opioid sparing approach to analgesia (think regional technique, multimodal analgesia)
(iv) Care with other sedatives
(v) Supplemental oxygen as indicated
(vi) Optimise respiratory function

What is the appropriate location of care following discharge form recovery for this woman?

Given Sally's condition and potential for deterioration, she requires a greater level of nursing and medical supervision than the general ward can provide. She should be transferred to a high dependency level bed in HDU/ICU.
She should wear her CPAP mask when sleeping and supplementary oxygen (as standard with her PCA morphine). Appropriate multimodal analgesia and aggressive physiotherapy are indicated.